

Table 3: Out-of-town check-ins of three selected users from test set.

User	Out-of-town check-ins
u_1	Scenics A, Scenics B, Art Gallery, Shopping Center A
u_2	Shopping Center B, Exhibition Center, Life Plaza, Shopping Center C, Hotel A
u_3	Enterprise, Hotel B, Hotel C

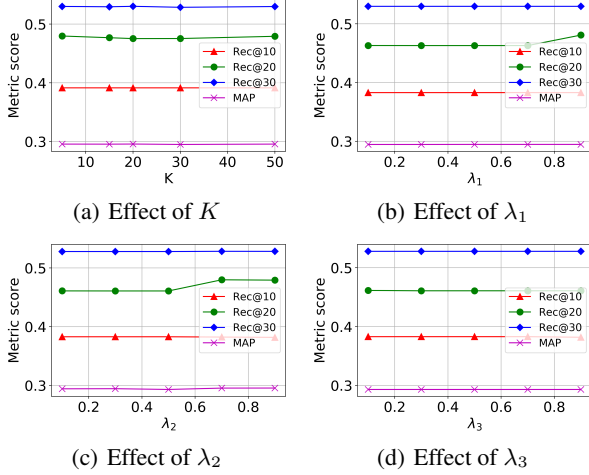


Figure 4: The impacts of inherent intention number (K) and different loss function weights (λ_1 , λ_2 and λ_3) on **SH**→**HZ** dataset w.r.t. the recommendation performance.

Parameter Sensitivity. We report the influence of the number of inherent travel intentions, i.e. K , and the impacts of λ_1 , λ_2 and λ_3 in loss functions, respectively. We only demonstrate the results on **SH**→**HZ** dataset. The results are similar on the other two datasets. Besides, to evaluate the impacts of parameters λ_1 , λ_2 and λ_3 on the recommendation performance, we first vary λ_1 while setting $\lambda_2 = \lambda_3 = \frac{1-\lambda_1}{2}$, and we use the similar strategy for evaluating λ_2 and λ_3 , respectively.

As shown in Fig. 4(a), the scores are relatively stable as K increases. The reason behind is that the user-specific intention is a summary of all generic intention embeddings, which makes the overall performance insensitive to the number of generic inherent intentions. Moreover, from Figs. 4(b) to 4(d), we can observe that, with increasing λ_1 , λ_2 and λ_3 from 0.1 to 0.9, both the Recall and MAP metrics are stable in general with an exception that the Rec@20 is slightly better with larger λ_1 and λ_2 .

Related Work

Out-of-town POI recommendation attempts to provide a list of POIs out-of-town users are interested in (?). To tackle the out-of-town recommendation problem, many informative features had been investigated, including user pref-

erence (?), geographical influence of out-of-town check-ins (??), and social influence (?). One important phenomenon that out-of-town recommendations should not neglect is user’s interest drift, i.e. user’s out-of-town check-ins are not aligned to user’s home-town check-in preference. Some studies have paid attentions to make out-of-town recommendation by taking into account user’s interest drift (???). Most of them take the textual reviews as input with employing the topic model. However, the data sparsity issue is getting worse when utilizing the textual content related to POIs and users. Besides, as a number of POIs in out-of-town check-ins relate to tourism, there also exist researches focusing on out-of-town tourism POI recommendation (???). Our work differentiates itself from previous works by comprehensively considering user’s preference, travel intention, geographical constraints and user interest drifts for out-of-town recommendation.

Topic models have been widely applied as generative models for different tasks (????). However, as the dimensionality grows, these methods are scant to perform fast and accurate inference. Recently, deep learning techniques and neural variational inference have accelerated the development of latent variable models (????). For example, (?) proposes a neural variational document model (NVDM) for text mining. Besides, Neural Topic Model (NTM) (?) was proposed to discover latent topics by variational inference. These methods provide us new opportunities to tackle topic discovery problem in a data driven fashion.

Concluding Remarks

In this paper, we studied the out-of-town recommendation problem via travel intention modeling. We proposed a data-driven framework TRAINOR to learn an out-of-town recommender by comprehensively taking user preference, interest drifts, travel intention and out-of-town geographical influence into account as a whole. To investigate user’s home-town preference, a G-GNN model was exploited. On the other hand, the user’s out-of-town preference was estimated in a collective manner and enriched through a geographical GCN. Afterward, we devised a preference transfer module to map home-town preference to out-of-town check-in behavior via an MLP. Moreover, to understand the user’s complex travel intention, we developed an NTM based user-specific travel intention discovery module. Finally, with jointly minimizing composite loss, TRAINOR can yield the learned recommender in an end-to-end fashion. Through extensive experiments on real-world datasets, we validated the effectiveness of TRAINOR quantitatively. A case study further validated the ability of TRAINOR to understand users’ travel intentions.

Acknowledgments

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